

National University AI Academic Assistant & Support Platform

Autonomous AI Conversational Assistant, Tracked Support Ticket Service & 24/7 Knowledge Enrichment Ecosystem

Project Mission & Executive Overview

National University (**nu.ac.bd**) is the largest higher-education affiliating university in Bangladesh, serving over 3 million students across 2,200+ affiliated colleges. Due to high query volumes during admissions, exam schedules, EMS portal logins, and certificate issuances, this enterprise AI platform was architected. It provides instant verified academic guidance and manages official, encrypted support tickets for departmental solvers.

⚡ Sub-Millisecond Response

High-frequency queries (greetings, admissions, exam routines) served in **0.018 ms** from in-memory cache.

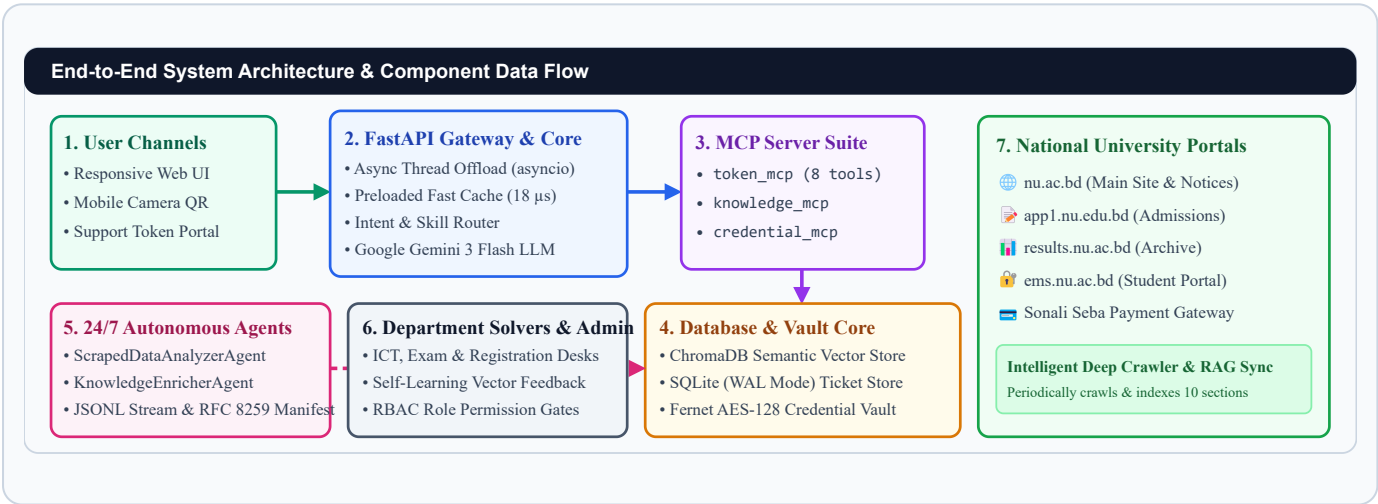
🛡️ Atomic Support Tokens

EMS lockout, form fill-up, and certificate issues resolved via **NU-2026-XXXXXX** tracked tickets.

🤖 24/7 Autonomous Agents

Continuously crawl, extract key dates/fees, vectorize Q&As, and update ChromaDB vector storage in real time.

1. Full System Architecture & Data Flow Diagram



2. Core Technologies & Architecture Principles

Technology	Version / Component	Role & Engineering Mechanism
FastAPI Backend	v0.115+ (Python 3.13)	Non-blocking asynchronous REST API gateway. Uses <code>asyncio.to_thread</code> to offload heavy disk I/O and generative AI calls, preventing event-loop latency.
Google Gemini 3 Flash	gemini-3-flash-preview	High-speed bilingual generative LLM core. Synthesizes contextual responses with verified official circular citations and active hyperlinks.
ChromaDB Vector Store	gemini-embedding-001	Persistent vector database for academic notices and solved support cases. Includes resilient pseudo-embedding fallback during quota limits.
MCP Server Suite	5 Dedicated Servers	Model Context Protocol tool suite adhering to Antigravity & Anthropic standards (<code>token_mcp</code> , <code>knowledge_mcp</code> , <code>document_mcp</code> , <code>credential_mcp</code> , <code>enrichment_mcp</code>).
SQLite Relational Core	WAL Mode Enabled	ACID relational engine for support tickets, credentials, crawler pages, and audit logs. Configured with Write-Ahead Logging for high write concurrency.
Fernet AES-128 Vault	CBC + HMAC-SHA256	Protects student service passwords with AES-128 encryption and PBKDF2 hashing (100,000 iterations). Plaintext passwords are never stored or logged.
Preloaded Fast Cache	In-Memory Trie (18 μs)	Preloads greetings (hi, hello), admissions criteria, routine FAQs, and SMS result formats into memory to respond in 0.018 milliseconds .

3. 24/7 Autonomous Multi-Agent Knowledge Enrichment Ecosystem

Agent Identifier	Core Responsibilities	Standard Output
ScrapedDataAnalyzerAgent	Scans crawled web pages & PDFs, extracts academic entities (dates, fees, degrees), and synthesizes bilingual Q&A pairs.	JSON Entities & QAs
KnowledgeEnricherAgent	Ingests Q&A pairs into ChromaDB vector store and updates in-memory fast caches for high-frequency user questions.	ChromaDB Ingestion
KnowledgeProvenanceAgent	Maintains append-only audit trail so peer AI models (Claude, Codex, Subagents) can synchronize and act on new knowledge.	JSONL Stream & Manifest

💡 Self-Learning Vector Feedback Loop

When an authorized departmental officer marks a support ticket as **SOLVED** with verified resolution instructions, the system anonymizes student identifiers and embeds the problem-solution pair into ChromaDB. Future students facing the same problem instantly receive the verified answer from the AI!

4. 0-Level Beginner User Operations Playbook

1. Ask the AI Assistant

Type any inquiry in natural English or Bengali (e.g. "Honours admission date" or "Forgot EMS password"). Receive instant answers with verified portal links.

3. Live Status Tracking

Click **Check Token** to see real-time solver updates (**PENDING** → **PROCESSING** → **SOLVED**) and download official resolution PDFs.

2. Apply for Support Token

For account or exam issues, click **Token Service**, choose your service, and submit. Receive an atomic tracking ID (e.g. **NU-2026-000140**).

4. Smartphone QR Code Access

Click **Mobile QR** to display a QR code. Scan with any mobile camera to access the full platform instantly without downloading applications.

5. Support Token Service Directory & Solver Desks

Service Code	Service Scope & Description	Responsible Desk	Resolution SLA
EMS	EMS Student Portal Login & Account Lockout Assistance	ICT Support Desk	24-48 Hours
FORM_FILLUP	Exam Form Fill-up & College Fee Verification Issues	Exam Controller Wing	12-24 Hours
RESCRUTINY	Result Re-check / Board Challenge Application Tracking	Evaluation Wing	3-7 Business Days
CERTIFICATE	Original & Provisional Certificate Processing & Release	Certificate Section	2-5 Business Days
MARKSHEET	Academic Transcript & Grade Sheet Corrections	Registrar Department	2-4 Business Days
TC / CORRECTION	College Migration Transfer & Student Profile Corrections	Registration Cell	5-10 Business Days

6. Strategic Multi-Phase Expansion Roadmap

Phase	Timeline	Planned Features & Engineering Capabilities
Phase 2 (Upcoming)	Q4 2026	Bangla Voice AI Assistant: Bidirectional real-time voice streaming for rural and visually impaired students.
Phase 3	Q1 2027	Automated SMS & WhatsApp Alerts: Instant notifications to student smartphones upon ticket resolution or urgent circulars.
Phase 4	Q2 2027	Robotic Digital Certificate Dispatch: University central database integration for instant digital-signature e-certificates.
Phase 5	Q3 2027	Federated Sub-Agent Clusters: Regional sub-agent pods for Dhaka, Chittagong, Rajshahi, and other administrative divisions.